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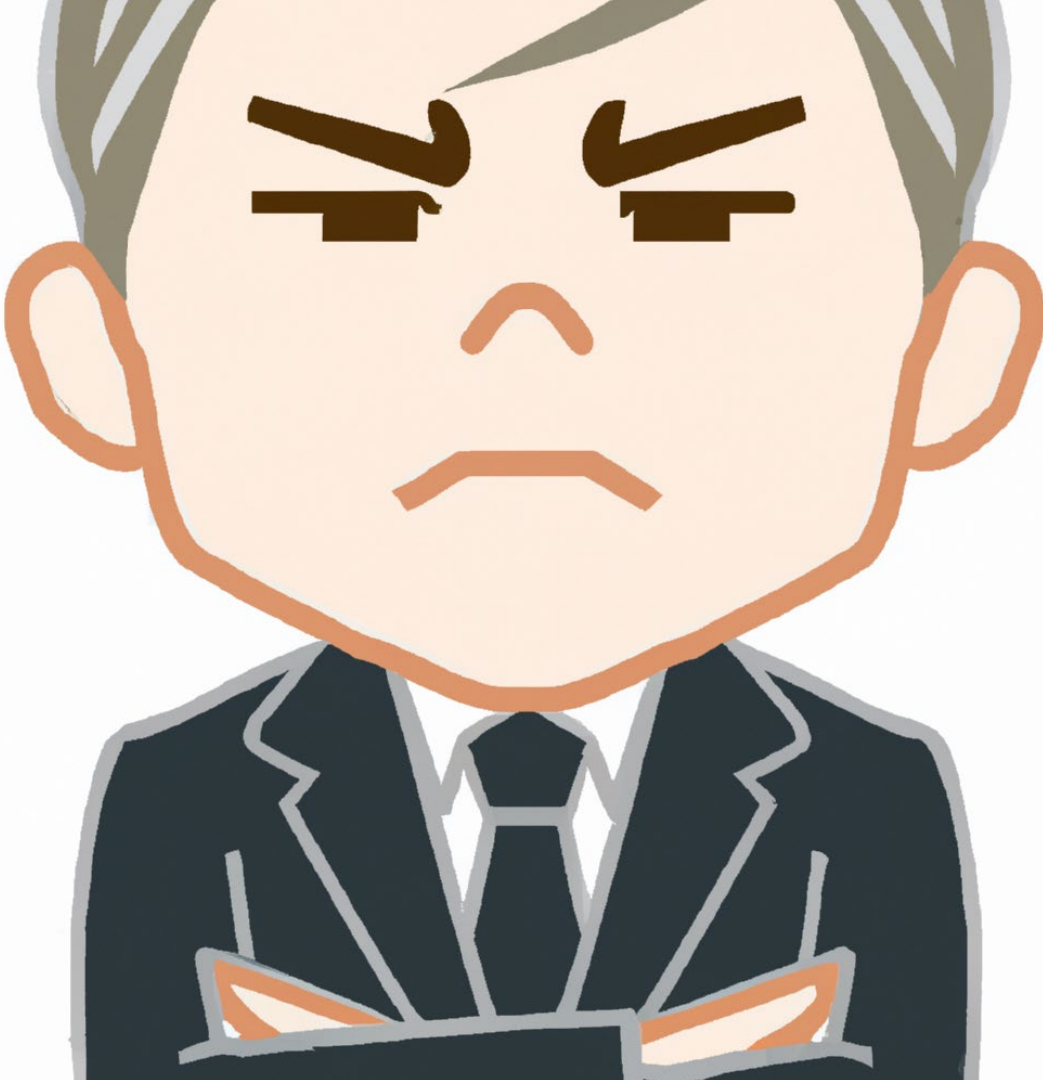
Norwegian University of Science and Technology

TDT4237: Lecture 1

Security principles



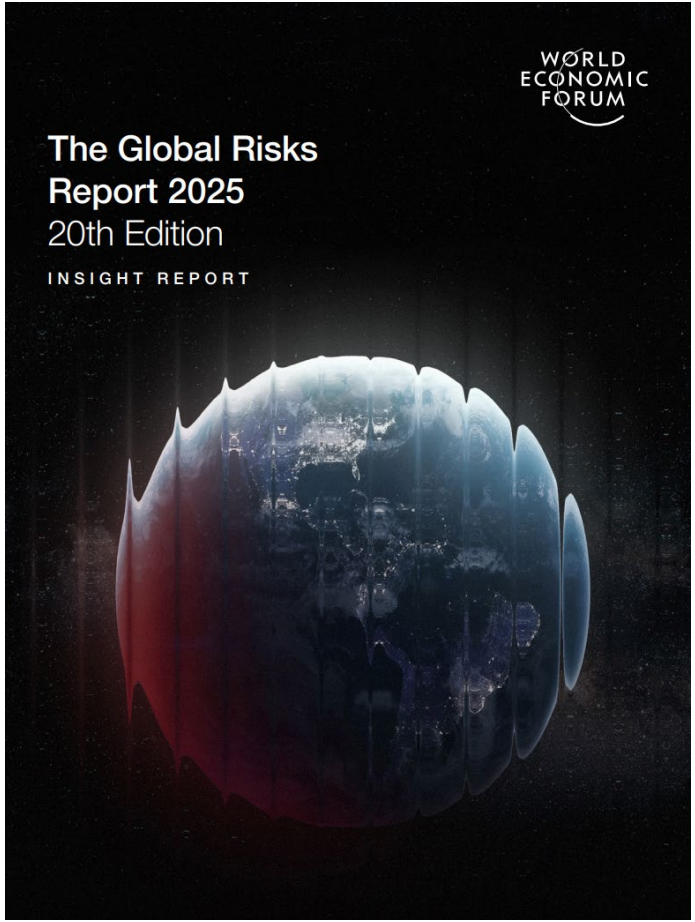
Outline

- Why security matters?
- Some examples of threats or attacks
  - Web defacement
  - Infiltration, control hijacking
  - Phishing
  - Data theft or loss
  - Denial of service
  - Ransomware
- Basic security goals
- Security guidelines

Why security matter?

- 81,6% of organizations suffered from a successful cyberattack last year. 28,7% had six or more.
- 92,1% of organizations have challenges related to attack surface management

2025 Cyberthreat Defense Report
https://co-c.net/ressources/papers/2025-CDR-Report-v1.0.pdf



Risk categories	2 years	
Economic	1st	Misinformation and disinformation
Environmental	2nd	Extreme weather events
Geopolitical	3rd	State-based armed conflict
Societal	4th	Societal polarization
Technological	5th	Cyber espionage and warfare
	6th	Pollution
	7th	Inequality
	8th	Involuntary migration or displacement
	9th	Geoeconomic confrontation
	10th	Erosion of human rights and/or civic freedoms

Source

World Economic Forum Global Risks Perception Survey 2024-2025.

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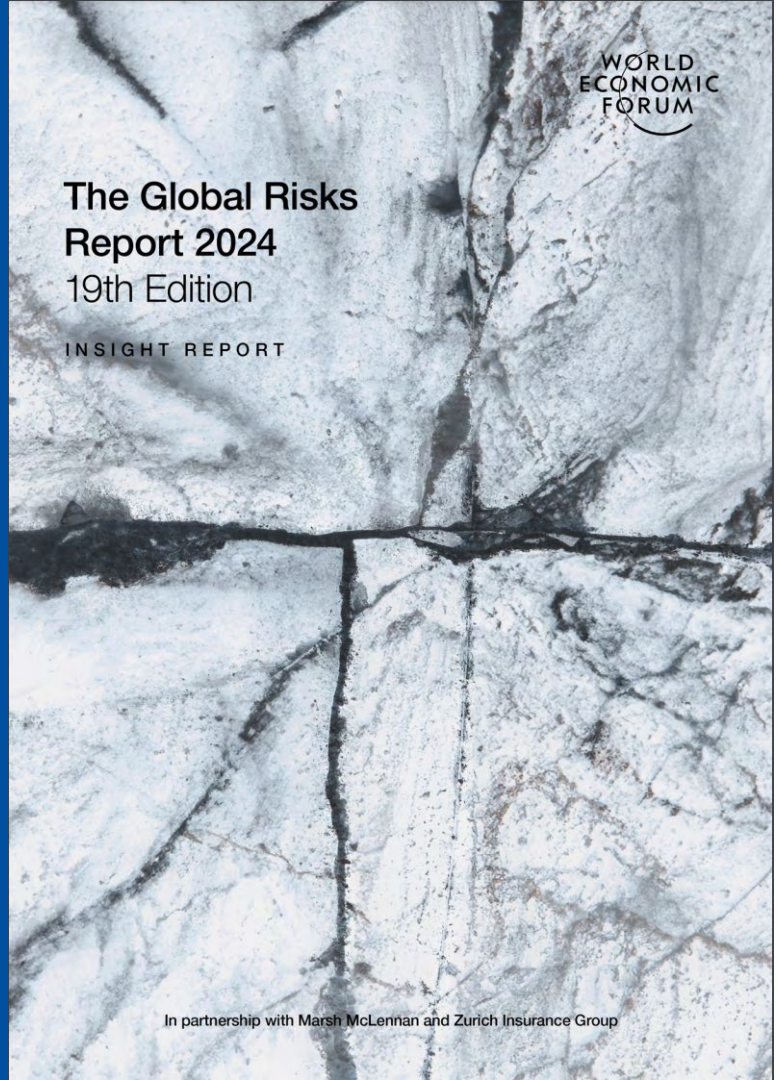

Risk categories

- Economic
- Environmental
- Geopolitical
- Societal
- Technological

2 years

1st	Misinformation and disinformation
2nd	Extreme weather events
3rd	Societal polarization
4th	Cyber insecurity
5th	Interstate armed conflict
6th	Lack of economic opportunity
7th	Inflation
8th	Involuntary migration
9th	Economic downturn
10th	Pollution

Source

World Economic Forum Global Risks Perception Survey 2023-2024.

Some examples of threats and attacks

Web defacement

- Replace legitimate pages with illegitimate ones

Read more: http://news.softpedia.com/news/IBM-Developer-Community-Website-Defaced-177457.shtml#ixzz4F7wmPpr6

Web defacement (cont’)

The Federal Depository Library Program (fdlp.gov)

Following the U.S. drone strike in Iraq in 2020

Image Source: Dailymail.co.uk

Infiltration, control hijacking

- Android: DroidDream Malware (2011)
- Infected 58 apps on Android marketplace
- 260,000 downloads in 4 days
- Sent premium-rate SMS message at night

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‘I lost 25 pounds in 20 days’: what it’s like to be on the frontline of a global cyber-attack

The security chief of SolarWinds reflects on the Russian hack that exposed US government agencies – and the heart attack he suffered in the aftermath

Tim Brown, the chief information security officer of SolarWinds, says it felt like the world was ‘on fire’ during the 2020 hack. Photograph: Sean Davey/The Guardian

Josh Taylor *Technology reporter*

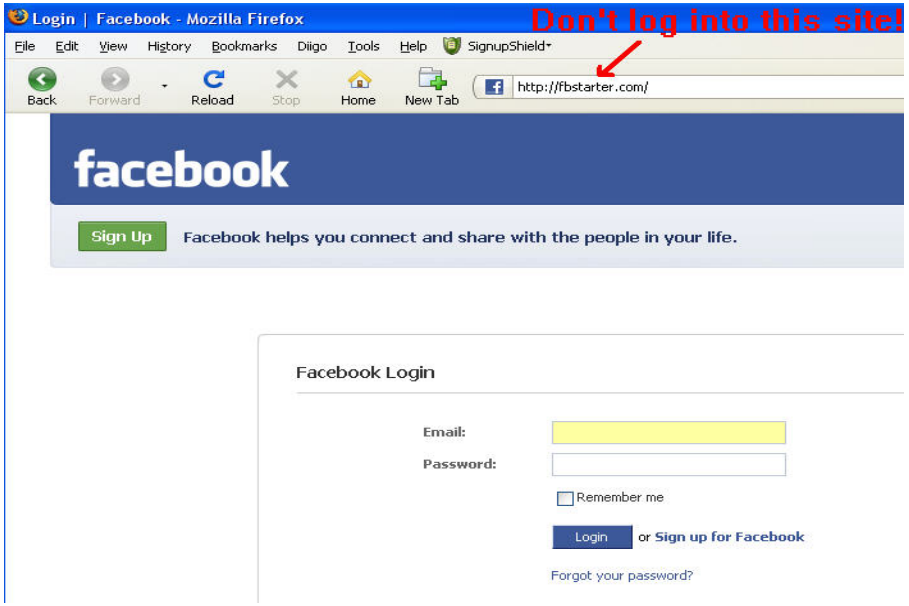
Sat 18 Oct 2025 21.00 CEST

- 2019: The Russian intelligence group Midnight Blissard got access to the SolarWinds network
- 2020: Malicious code inserted into the Orion network software
- SolarWinds distributed the Orion updates to 18,000 customers, including U.S. federal agencies
- Enabled long-term espionage (6 months)

https://www.theguardian.com/technology/2025/oct/19/global-cyber-attack-russian-hack-solarwinds-stress-health

Phishing

- Spoofed site that looks real



Read more: New Phishing Attack Spreading On Facebook. This Time From Fbstarter (2009).
https://techcrunch.com/2009/04/30/new-phishing-attack-spreading-on-facebook-this-time-from-fbstarter/

Phishing (cont’)

- Lure user through, e.g., phishing emails or SMS

Data theft / Data loss

May 2016:

Another Day, Another Hack: 117 Million LinkedIn Emails And Passwords

January 2023:

Twitter hacked, 200 million user email addresses leaked, researcher says

https://www.vice.com/en/article/78kk4z/another-day-another-hack-117-million-linkedin-emails-and-password
https://www.reuters.com/technology/twitter-hacked-200-million-user-email-addresses-leaked-researcher-says-2023-01-05/

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Reuters

World Business Markets Sustainability Legal Breakingviews More

Hacker claims breach of US location tracking company Gravy Analytics

By Raphael Satter and A.J.

January 9, 2025 12:45 AM GMT

Companies

Gravy Analytic

Redsense Med

ASHBURN, Virginia, Jan 8 (

location tracking firm Gravy

It is not clear exactly how ar

and screenshots uploaded e

carried a claim that the com

GRAFIKK: HENRIK LIED / NRK

Denial of service (DoS)

- Flood server with packets
- Cause server to drop legitimate packets
- Make service unavailable

Oct. 2012

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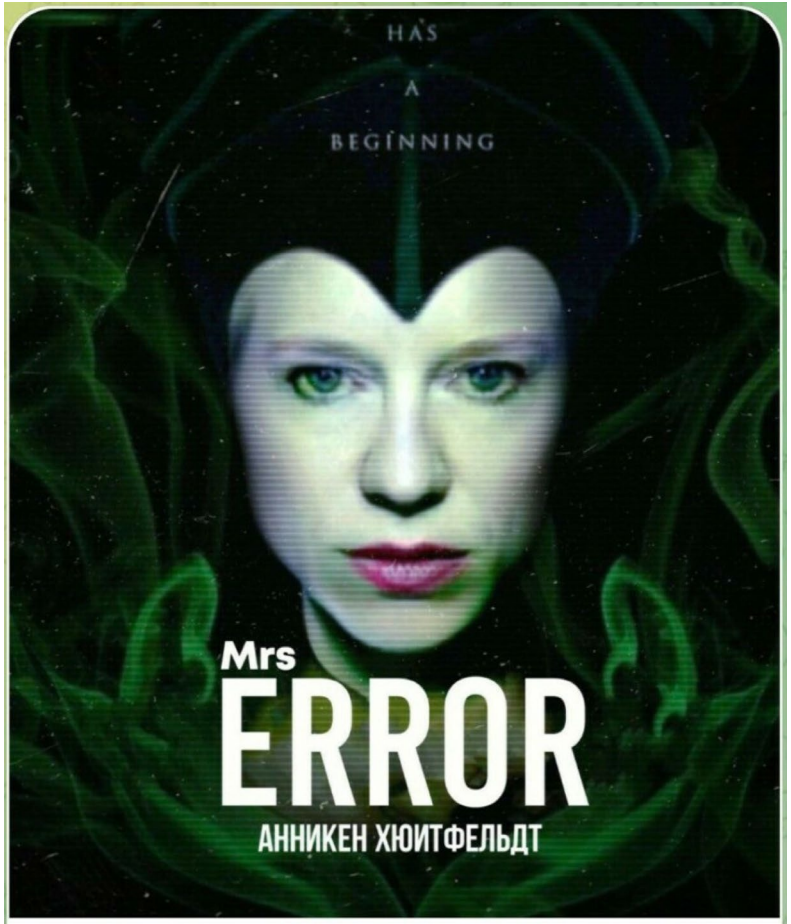
June 2022:



ЛЕГИОН - КИБЕР СПЕЦНАЗ РФ

🌤 Доброе утро Норвегия! ❗ Всем отрядам к бою ❗ ⚡ L7...

Казалось бы, причем тут Норвегия?

"Норвежские власти отклонили заявку России на пропуск грузов для российских поселков на Шпицбергене через единственный пропускной пункт на российско-норвежской границе Стурскуг, сообщили в МИД страны.
«Конкретная заявка на получение разрешения на перевозку была отклонена 15 июня 2022 года», – сообщило телеканалу NRK министерство, передает РИА «Новости».

В апреле глава МИД Норвегии Анникен Хюитфельдт заявила о решении закрыть для проезда грузовиков из России пункт пропуска Стурскуг на границе. Также было решено ввести запрет для захода судов из России в порты, кроме рыболовных"

2.2K изменено 09:44

Ransomware

Dear Customer:

It is time to pay for your software lease from PC Cyborg Corporation. Complete the INVOICE and attach payment for the lease option of your choice. If you don't use the printed INVOICE, then be sure to refer to the important reference numbers below in all correspondence. In return you will receive:

- a renewal software package with easy-to-follow, complete instructions;
- an automatic, self-installing diskette that anyone can apply in minutes.

Important reference numbers: A5599796-2695577-

The price of 365 user applications is US$189. The price of a lease for the lifetime of your hard disk is US$378. You must enclose a bankers draft, cashier's check or international money order payable to PC CYBORG CORPORATION for the full amount of $189 or $378 with your order. Include your name, company, address, city, state, country, zip or postal code. Mail your order to PC Cyborg Corporation, P.O. Box 87-17-44, Panama 7, Panama.

Press ENTER to continue

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CYBERSECURITY

Grounded. Why the European airports cyber incident is a wake-up call

Sep 25, 2025

An urgent wake-up call on the need to build cyber resilience across all critical infrastructure.

Image: REUTERS/Isabel Infantes/File Photo

https://www.weforum.org/stories/2025/09/european-airports-cyber-incident-critical-infrastructure/

Discussion (5 minutes)

- What do you think these companies could have done to prevent these attacks?

Basic security goals

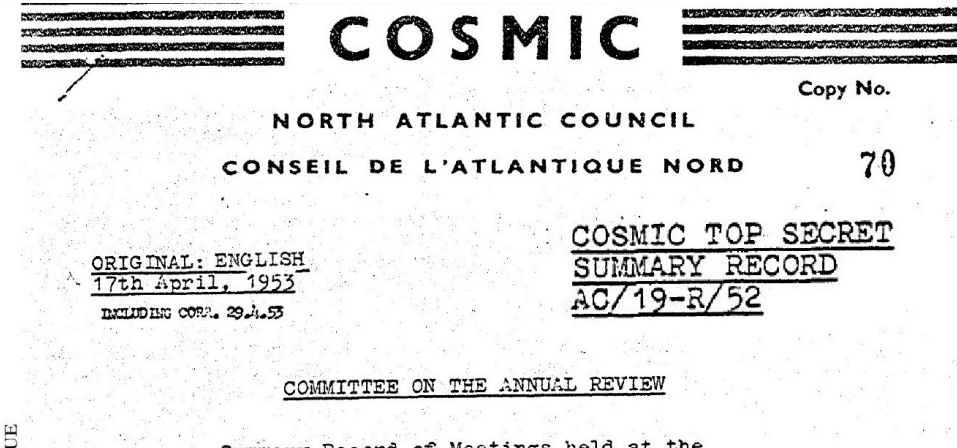
- **C**onfidentiality
- **I**ntegrity
- **A**vailability

- Privacy
- Accountability
- Non-Repudiation

Confidentiality

- Keep something secret
  - Communication
  - Data on storage (at rest)

- Typically accomplished with
  - Cryptography
  - Authentication
  - Authorization
  - Sealed envelopes
  - Etc.



COSMIC

Copy No.

NORTH ATLANTIC COUNCIL

CONSEIL DE L'ATLANTIQUE NORD

70

COSMIC TOP SECRET
SUMMARY RECORD
AC/19-R/52

ORIGINAL: ENGLISH
17th April, 1953
INCLUDING CORR. 29.4.53

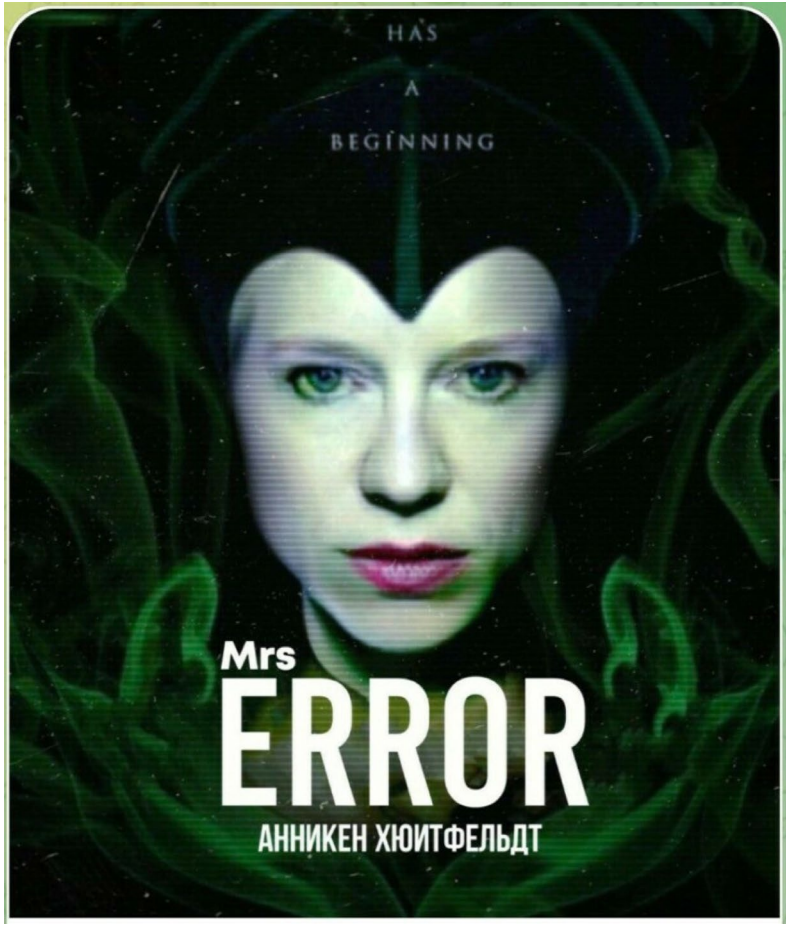
COMMITTEE ON THE ANNUAL REVIEW

Integrity

- Data integrity = No corruption
- Control integrity = No control hijacking
- Different from confidentiality
  - Confidentiality: who can read the message
  - Integrity: who can write the message
- Sometimes accomplished with
  - Message/data hashing

Availability

- System uptime
- System response time
- Free storage



Privacy

- Right to be left alone
- Different from confidentiality
  - Confidentiality: the secret of business information
  - Privacy: the protection of personal information

Accountability

- Logging and audit trails
- Accomplished by
  - Secure timestamping
  - Data integrity in logs and audit trails

Non-Repudiation

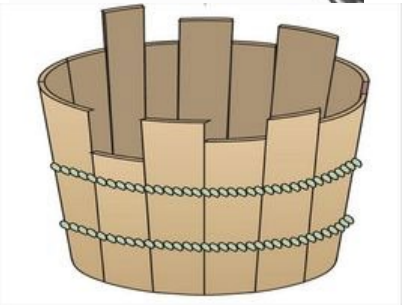
- Two parties cannot deny that they have interacted with each other
- A trusted 3rd party can be used
  - E.g., Alice wants to prove to Trent that she did a transaction with Bob
- Generate evidence/receipts (digitally signed statements)

Security guiding principles

1. Secure the weakest link
2. Practice defense in depth
3. Fail securely
4. Compartmentalize
5. Be reluctant to trust
6. Follow the principle of least privilege
7. Keep it simple
8. Promote privacy
9. Remember that hiding secrets is hard
10. Use your community resources

1. Secure the weakest link

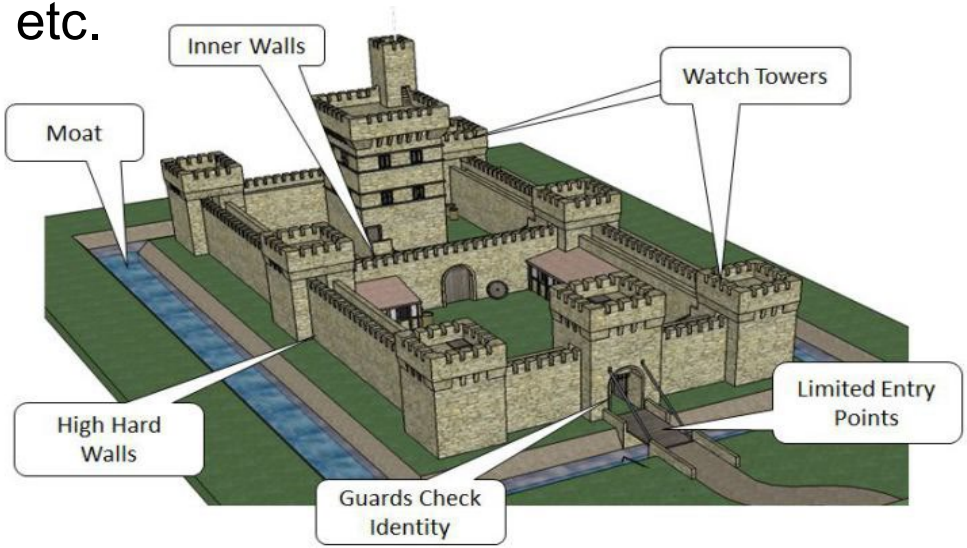
- ***Information security is as strong as its weakest link***
- The attacker only needs to find one flaw
- Designers have to try and cover all possible flaws
- Common weak links
  - Weak passwords
  - People: social engineering attacks, internal attacks
  - Poor software



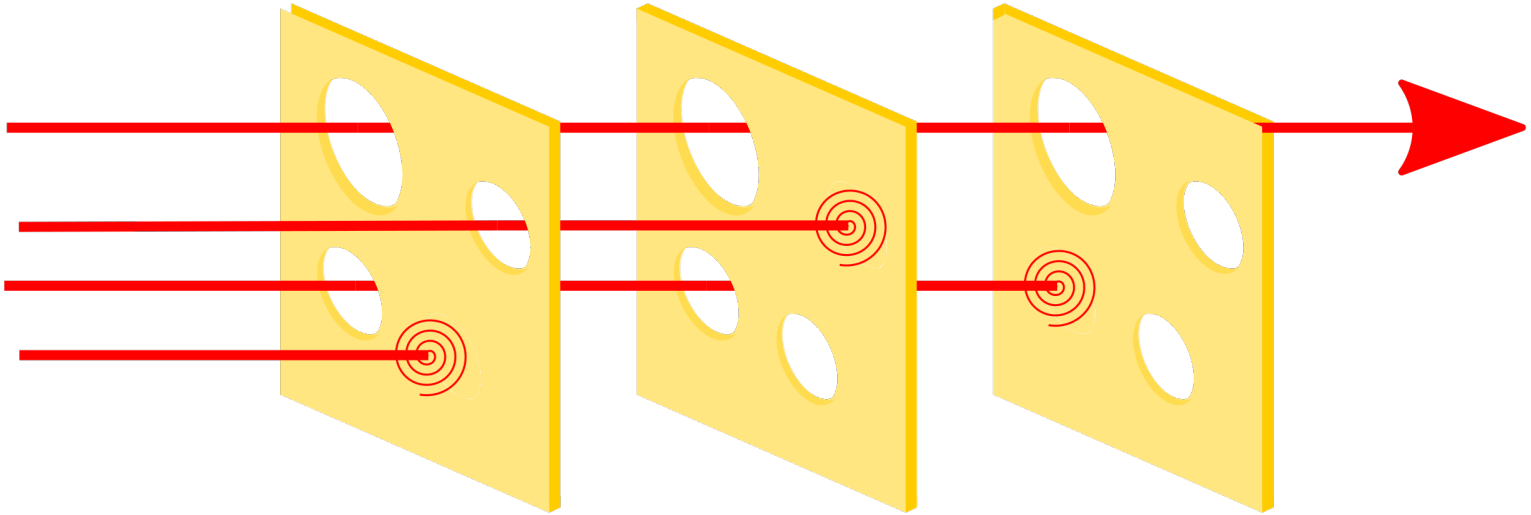
Cannikin Law

2. Practice defense in depth

- Layers of defense
- Do not rely on one-shot security
  - E.g., firewall + authentication + authorization + cryptography, etc.



Swiss cheese model



https://en.wikipedia.org/wiki/Swiss_cheese_model#/media/File:Swiss_cheese_model_textless.svg

2. Practice defense in depth (cont’)

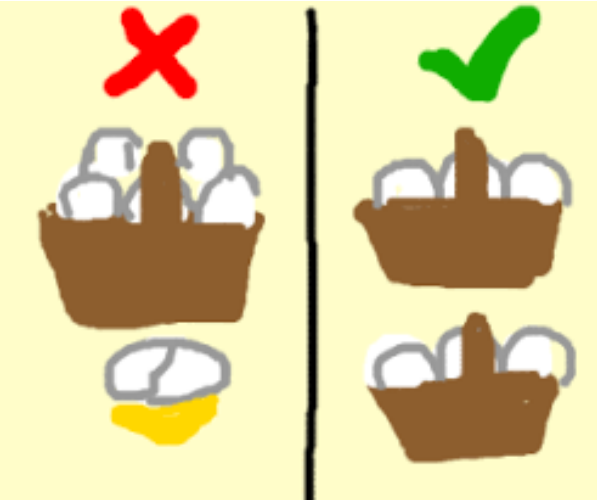
- Prevention
- Detection
- Containment (e.g., emergency response plan)
- Recovery

3. Fail securely

- Expect failure of security features
  - Exception of a security control itself
    - E.g., if a firewall fails, let no traffic in
  - Other exceptions can cause a security feature not to be invoked
    - E.g., when the line to the credit card company is down, no online credit card authentication, still allow transactions?

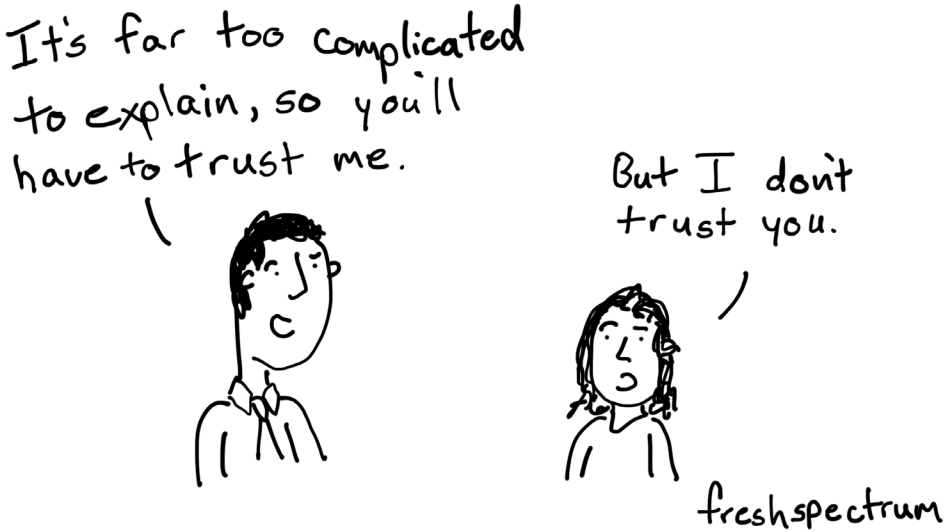
4. Compartmentalize

- Separate something (e.g., code) into parts
- Don’t mix those parts, e.g.:
  - Separate network into different zones
  - Run the sensitive application on separate computers



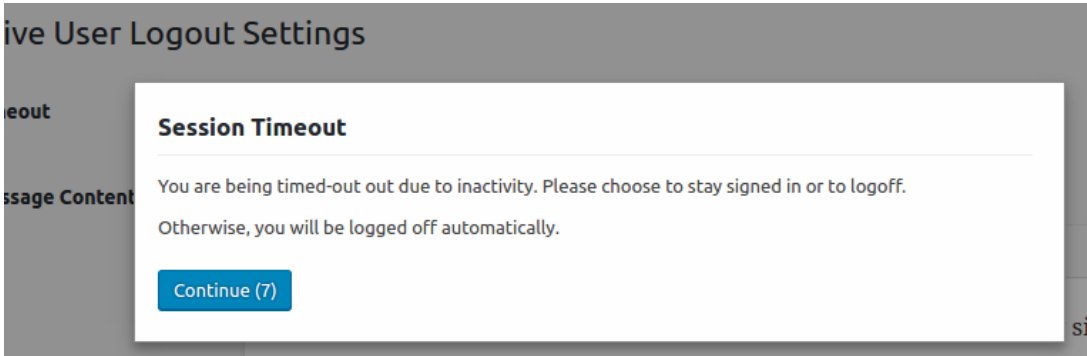
5. Be reluctant to trust

- Skepticism is always good
  - Don't trust any code library
- Should not trust or assume the validity of user inputs
  - E.g., SQL injection attack



6. Follow the principle of least privilege

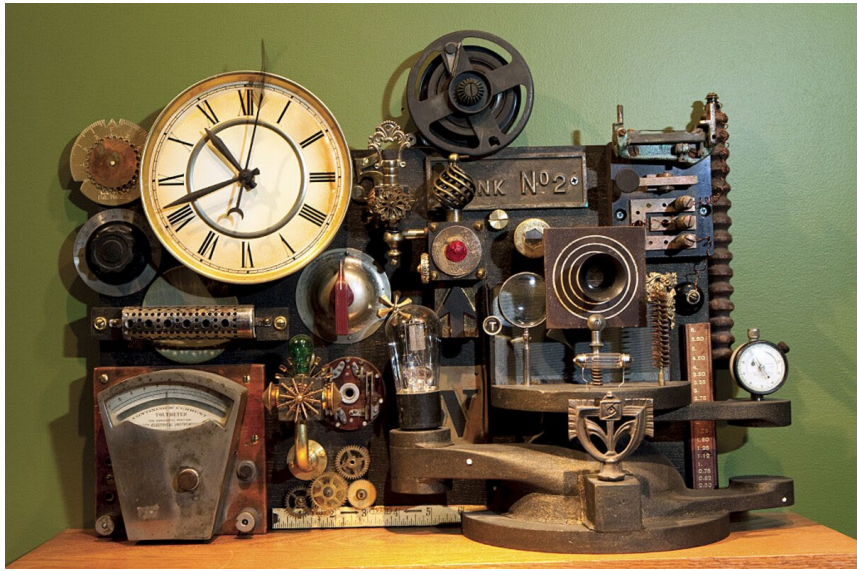
- Minimum access necessary to get the job done
  - E.g., only system admin can read and modify system files
  - E.g., web server can read, but cannot modify .html file
- Minimum amount of time necessary
  - E.g., after a user is inactive for a while, the system logs out the user automatically



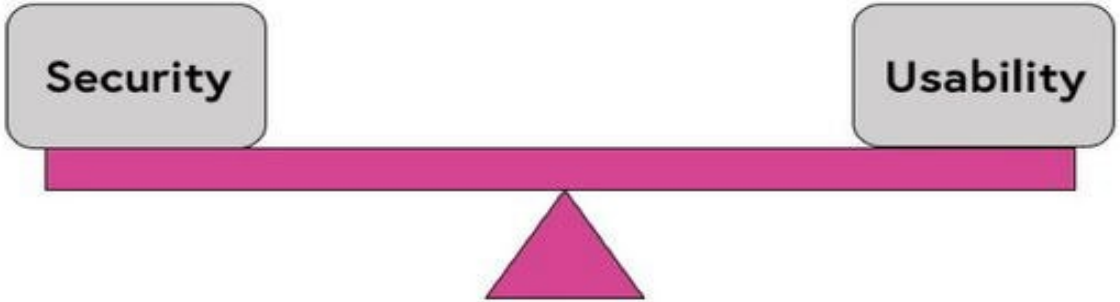
7. Keep it simple

- Make systems simple
  - Reduce attack surface
  - Less functionality = less security exposure
  - All unnecessary features/functions off
  - Close unnecessary ports

https://www.csmonitor.com/The-Culture/Arts/2010/1229/Steampunk-The-new-Goth



7. Keep it simple (cont')



- Tradeoff: relative security benefit at only slight inconvenience

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PIN-kode fra SMS
Vi vil nå sende deg en valideringskode på SMS, vennligst bekreft at dette telefonnummeret er korrekt: +474
BEKREFT
Eller legg inn korrekt telefonnummer ved å trykke her

8. Promote privacy

- Do not to compromise the privacy of the user
- “nice-to-have” VS necessary information

9. Remember that hiding secrets is hard

- “Security by obscurity”
  - Maybe necessary, but not sufficient
- Attackers can probe for weaknesses
- Open vs. Close Source
  - A business decision, not a security one

10. Use your community resources

- Websites and sources of information
- “Known threats" and vulnerabilities
- E.g.,
  - Common vulnerabilities and Exposures
    - https://cve.mitre.org/
  - National vulnerability database
    - https://web.nvd.nist.gov/view/vuln/search
    - https://web.nvd.nist.gov/view/vuln/statistics
  - OWASP Top 10
    - https://www.owasp.org/index.php/OWASP_Top_Ten_Project

Next lecture…